Temporal Superposition in GPT-2: How Time is Encoded and Distributed Across Layers

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Abstract

We investigate how GPT-2 Small encodes temporal concepts in its residual stream, combining PCA of raw activations with a layer-wise sparse autoencoder (SAE) ablation scan across all 12 layers. We study both linear temporal representations using a 100-prompt historical year corpus and circular representations using three cyclic concept datasets: days of the week, months of the year, and days of the month.

For linear historical time, the temporal axis is strongly present ($|r|_{\text{base}} \approx 0.85$–0.92) throughout the network, but transitions from concentrated to distributed with depth: at layers 0–1 a single SAE feature accounts for a drop of ≈ 0.43 in temporal correlation when ablated, whereas by layer 7 no individual feature is load-bearing. Removing BC/AD surface tokens collapses the layer-1 signal to near zero and shifts peak concentration to layers 4–5, after which both corpora converge to the same distributed superposition regime — demonstrating that surface token identity can dominate early-layer probing signals in ways that obscure genuine semantic encoding.

For cyclic concepts, we replicate the circular PC2/PC3 geometry of prior work for days of the week and months of the year, and extend it to days of the month ($r_{\text{angular}} = 0.495$), a 31-label concept not previously studied in this context. A layer-wise ablation scan of circular signal strength reveals that the two well-established cyclic concepts follow opposite depth trajectories: days of the week are encoded strongly from early layers but collapse at layer 11 ($r_{\text{angular}} = 0.266$), while months of the year build progressively and peak at layer 11 ($r_{\text{angular}} = 0.793$). Neither concept shows the monotonic decline in feature concentration seen for linear time, suggesting cyclic representations require active, layer-by-layer maintenance rather than a single construction phase.

1 Introduction

Large language models encode rich temporal knowledge, yet how that knowledge is geometrically organised inside the residual stream remains a topic of research. Temporal concepts are an appealing target for mechanistic interpretability because they carry verifiable ground-truth structure — linear ordering for historical time, circular ordering for cyclic concepts such as days and months — enabling precise geometric tests rather than purely behavioural probes.

We apply PCA and a layer-wise SAE feature ablation scan to GPT-2 Small's residual stream across all 12 layers, studying both a 100-prompt historical year corpus and three cyclic concept datasets (days of the week, months of the year, days of the month). The ablation scan zeros individual SAE features and measures the drop in temporal signal, distinguishing *concentrated* representations (one feature is load-bearing) from *distributed superposition* (signal is spread across many features). Our results reveal a clean concentrated-to-distributed transition for linear time, a surface-token confound that masks genuine semantic encoding in early layers, and contrasting depth trajectories for the two well-studied cyclic concepts.

2 Related Work

TODO

3 Background

3.1 Residual Stream

Transformer models process tokens by passing representations through a sequence of attention and MLP layers. These layers write incremental updates into a shared vector called the residual stream. Formally if $\mathbf{x}^{(\ell)} \in \mathbb{R}^d$ denotes the residual stream at layer ℓ, each layer computes $\mathbf{x}^{(\ell+1)} = \mathbf{x}^{(\ell)} + f^{(\ell)}$, where $f^{(\ell)}$ is the combined attention and MLP transformation. This additive structure means the residual stream accumulates information across depth, and that representations at any layer can be read as a type of superposition of all contributions made so far. In this work we probe the residual stream of GPT-2 Small Radford et al. [2019], a 117 million parameter transformer with a hidden dimension of d = 768.

3.2 Superposition

A naive expectation is that each dimension of the residual stream encodes one interpretable feature. In practice, the number of concepts a model must represent exceeds its hidden dimension d ($\mathbb{R}^{768}$ in this paper). As a result, models exploit the fact that high dimensional spaces can accommodate many nearly-orthogonal directions. This phenomenon is called superposition and it allows a model to store a large amount of features at the same time. Features in superposition are distributed across many neurons.

3.3 Principal Component Analysis

Principal Component Analysis (PCA) finds a set of orthogonal directions, called principal components, that capture the maximum variance in a dataset. Given a centred data matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$, PCA computes the eigendecomposition of the covariance matrix $\mathbf{C} = \frac{1}{N}\mathbf{X}^\top \mathbf{X}$. The eigenvectors of $\mathbf{C}$ define the principal component directions, and the corresponding eigenvalues quantify the variance explained along each direction. The first principal component (PC1) is the eigenvector with the largest eigenvalue — the direction of greatest variance in the data. If a model encodes a concept as a linear axis in activation space, that axis should dominate the variance structure and align with PC1. Secondary structure, such as the circular geometry of cyclic concepts, tends to emerge in PC2 and PC3 once the dominant linear source of variance has been accounted for by PC1.

3.4 Sparse Autoencoders

Sparse autoencoders (SAEs) offer a way to recover the underlying precise features from an ambiguous residual stream Cunningham et al. [2024]. An SAE consists of an encoder that projects the d-dimensional activation into a much larger latent space $\mathbb{R}^{d_{\text{SAE}}}$ with $d_{\text{SAE}} \gg d$, followed by a ReLU that enforces sparsity and a linear decoder that reconstructs the original activation. Training minimises reconstruction error subject to an L_1 penalty on the latent activations, encouraging each input to be explained by only a small number of active features at a time. The resulting latent dimensions have been shown to correspond to interpretable, monosemantic concepts Cunningham et al. [2024], Bloom [2024].

4 Method

4.1 Model and SAE

We do all experiments on GPT-2 Small Radford et al. [2019], a decoder-only transformer with 12 layers, 12 attention heads, and a residual stream dimension of d = 768, totalling approximately 117 million parameters. GPT-2 Small is a well-studied model in the context of mechanistic interpretability, making it a good model for probing internal representations.

To decompose residual stream activations, we use the suite of SAEs released by Bloom [2024], trained on GPT-2 Small's residual stream under the release identifier `gpt2-small-res-jb`. Each SAE is trained on the `hook_resid_pre` activation at its corresponding layer – that is, the residual stream state *before* the attention and MLP transformations of that layer have been applied. The SAE architecture uses a linear encoder with ReLU activation followed by a linear decoder, with an expansion factor of 64×, giving a latent dimension of $d_{\text{SAE}} = 49{,}152$. For each layer ℓ we load the corresponding SAE independently, allowing us to probe how feature representations evolve across depth.

All activations are extracted at the final token position of each prompt, as this is the position at which the model has processed the full input and is most likely to have integrated contextual information into a coherent representation.

4.2 Prompt Design

All prompts are short declarative sentences placing the concept of interest in natural linguistic context. The model processes each prompt without instruction or chat formatting, with activations extracted at the final token position.

Historical year corpus. The primary dataset consists of 100 prompts spanning approximately 2800 years of history, with 20 prompts per epoch category: `ancient_bc` (−776 to −31), `ancient_ad` (79–529), `medieval`, `early_modern`, and `modern` (through 1991). Each prompt is a single declarative sentence describing a well-known historical event at a specific year, for example *"The first Olympic Games were held in 776 BC"* or *"The Norman Conquest of England occurred in 1066"*. The balanced design ensures equal representation across epochs for ANOVA-based feature selection. BC and AD year ranges are non-overlapping by design — BC items span −776 to −31 and AD items span 79 to 529 — avoiding any ambiguity in numeric year assignment.

Surface token comparison. The superposition depth scan is run twice on the same 100 prompts to isolate the effect of BC/AD surface tokens. In the first scan, the 40 ancient prompts carry explicit era markers (e.g. *"...in 776 BC"*, *"...in 79 AD"*). In the second scan, era markers are stripped from all ancient prompts, leaving bare year numbers (e.g. *"...in 776"*). Medieval through modern prompts are word-for-word identical across both scans. This design ensures the only variable between the two conditions is the presence or absence of the tokens "BC" and "AD", with corpus size, event selection, year values, and category labels held constant.

Cyclic concept datasets. To examine whether cyclic temporal concepts are encoded with circular geometry, we construct three datasets. Days of the week ($N = 21$) and months of the year ($N = 36$) each represent their respective concepts across three prompt frames per item: a simple statement (*"Today is Monday"*), a wraparound sequential statement (*"The day after Sunday is Monday"*), and a positional statement (*"Monday is the first day of the working week"*). Days of the month ($N = 93$) extends this structure to all 31 days. The multi-frame design ensures geometric structure in the representations reflects the underlying concept rather than a single surface phrasing.

4.3 Temporal Axis Measurement: Linear Concepts

To quantify how strongly a layer encodes temporal order for historical year prompts, we measure the alignment between the model's internal representations and chronological position. For each layer l, we pass the year prompt activations through the pretrained SAE encoder and reconstruct them via the decoder:

$$\hat{x}_i = \mathbf{W}_{\text{dec}}\mathbf{a}_i \in \mathbb{R}^{768}$$

where $\mathbf{a}_i \in \mathbb{R}^{49152}$ is the sparse feature activation vector for prompt i. This reconstruction discards variance not captured by the SAE, meaning our metric reflects temporal structure as expressed through the SAE's feature basis specifically.

We apply Principal Component Analysis (PCA) to the $(N \times 768)$ reconstruction matrix, retaining a single component (PC1), and compute the absolute Pearson correlation between PC1 scores and numeric year values:

$$|r| = \left|\text{corr}\left(\text{PC1}(\hat{\mathbf{X}}),\ \mathbf{y}\right)\right|$$

where $y \in \mathbb{R}^N$ contains the numeric year for each prompt, with pre-common-era dates represented as negative integers. We use PC1 because it captures the direction of maximum variance in the reconstruction. If the model encodes temporal position as a linear axis, that axis should dominate the variance structure and align with PC1. The absolute value discards PCA's arbitrary sign convention. We refer to this quantity as the baseline temporal correlation $|r|_{\text{base}}$ at a given layer, and use drops in $|r|$ under feature ablation as the primary signal in our superposition analysis, described in Section 5.2.

4.4 Temporal Axis Measurement: Cyclic Concepts

For cyclic concepts – days of the week, months of the year, and days of the month – a linear correlation metric is inappropriate. Day 7 and day 1 are adjacent in the cyclic ordering but greatly separated under any linear projection, meaning Pearson r will be near zero even for a geometrically well-structured representation. We therefore adopt a circular measurement pipeline.

We retain five PCA components from the SAE reconstruction and fit an algebraic circle to the PC2/PC3 plane using the Coope(1993) least squares method Coope [1993]. PC1 is deliberately excluded as it captures non-cyclic variance (e.g. whether the text refers to a weekday versus a month); the circular structure of interest emerges in PC2 and PC3. For each prompt we compute its angular position relative to the fitted circle center (c_x, c_y):

$$\theta_i = \arctan 2\,(\mathrm{PC3}_i - c_y,\ \mathrm{PC2}_i - c_x)$$

We then compute the Fisher circular-linear correlation between angular position and the integer label $\mathbf{y}$. A standard Pearson correlation is inappropriate here because θ_i is a circular variable — day 7 and day 1 are adjacent in the cyclic ordering but nearly π apart in angle, causing Pearson r to underestimate the true correspondence. The Fisher metric instead projects the angular variable onto $\cos\theta$ and $\sin\theta$ and measures how well both components jointly predict the label, correctly treating the concept as periodic.

$$r_{\text{angular}} = \sqrt{\frac{r_{xc}^2 + r_{xs}^2 - 2r_{xc}r_{xs}r_{cs}}{1 - r_{cs}^2}}$$

where $r_{xc} = \mathrm{corr}(\cos\theta, \mathbf{y})$, $r_{xs} = \mathrm{corr}(\sin\theta, \mathbf{y})$, and $r_{cs} = \mathrm{corr}(\cos\theta, \sin\theta)$. This metric is bounded in $[0, 1]$ and correctly treats the label as periodic, with $r_{\text{angular}} = 1$ indicating that the label perfectly predicts angular position on the circle. For the ablation scan in Phase 5, drops in r_{angular} under single-feature ablation replace drops in $|r|$ as the signal metric, with all other aspects of the pipeline held constant.

4.5 Feature Selection for Ablation

Given the high dimensionality of the SAE feature space ($d_{\text{SAE}} = 49,152$), directly ablating all features is computationally unmanageable and statistically uninformative. We therefore apply a filtering procedure to identify a focused subset of features likely to be encoding temporal structure, which then forms the candidate set for the ablation scan.

We first filter to *active* features — those that fire on at least two prompts in the input set. For each active feature f_i we compute a one-way ANOVA F-score across grouping labels, where each group contains the feature activations for all prompts belonging to that group. For linear temporal concepts, groups correspond to historical epoch categories (ancient BC, ancient AD, medieval, early modern, modern); for cyclic concepts, groups correspond to the integer cycle labels (day-of-week 1–7, month 1–12, day-of-month 1–31). We retain the top $N_{\text{disc}} = 100$ features by F-score as the candidate set. This procedure is applied uniformly across all phases.

4.6 Superposition Depth Scan

To examine how temporal representations evolve across depth, we run a layer-wise ablation scan across all 12 layers of GPT-2 Small. At each layer l we load the corresponding pretrained SAE, extract residual stream activations at `hook_resid_pre`, and apply the feature selection procedure described in Section 4.5 to obtain the candidate feature set. We then establish a baseline temporal correlation $|r|_{\text{base}}$ by encoding the prompts through the SAE, reconstructing via the decoder, and computing the absolute Pearson correlation between PC1 and the numeric year values.

For each feature f_i in the candidate set, we zero its activation across all prompts simultaneously — setting column i of the ($N \times$ 49,152) feature activation matrix to zero — and recompute $|r|$ on the modified reconstruction. The drop is:

$$\Delta_i = |r|_{\text{base}} - |r|_{\text{abl},i}$$

A large Δ_i indicates that feature f_i is individually load-bearing for the temporal axis. A near-zero or negative Δ_i indicates negligible contribution or a suppressive effect whose removal improves temporal linearity. PCA is re-fit from scratch on each ablated reconstruction rather than projecting onto a fixed baseline direction, so Δ_i measures whether a temporal axis still exists after ablation, not merely whether the ablated reconstruction aligns with the original. The test is therefore conservative: a feature only registers as load-bearing if the remaining features cannot reorganise to compensate.

For the clean-prompt scan (Phase 4), BC/AD surface tokens are removed from all prompts, yielding a matched corpus of the same 100 historical years expressed without explicit era markers. The same feature selection and ablation procedure is applied, enabling a direct comparison of superposition depth under contaminated versus uncontaminated input conditions.

5 Experiments and Results

5.1 Temporal Structure in the Residual Stream

PCA of the raw residual stream at layer 4 reveals that GPT-2 Small organises historical prompts into three distinct clusters in the PC1/PC2 plane and four clusters in three dimensions (Figure 1): BC-era prompts, ancient AD-era prompts, medieval and early modern prompts, and modern prompts, accounting for 73.4% of variance across the top three principal components. PC1 (41.9%) separates ancient from modern prompts while PC2 (20.9%) separates BC from AD within the ancient period.

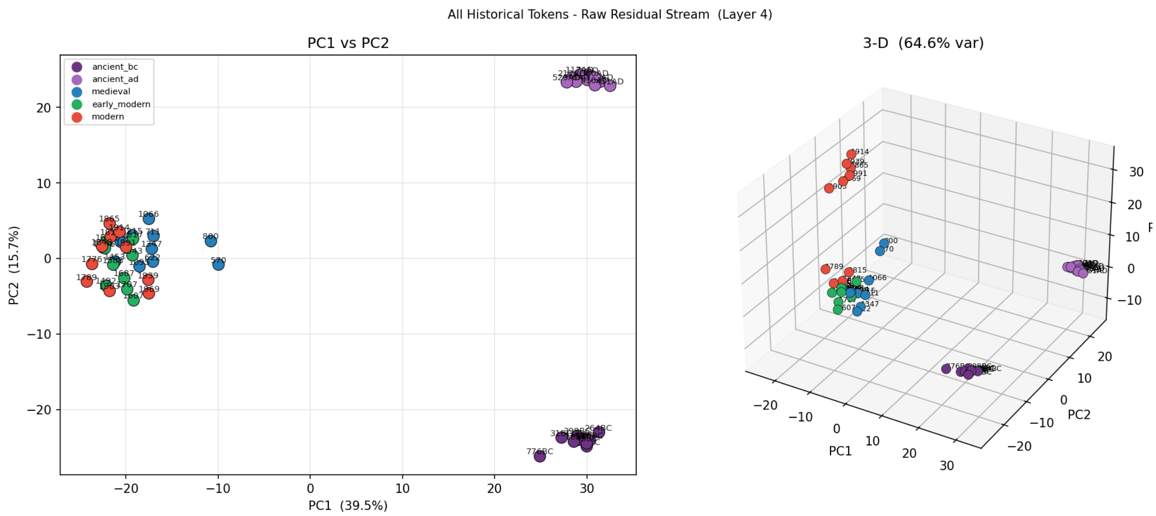

Figure 1: PCA of raw residual stream activations at layer 4. Left: PC1 vs PC2 coloured by epoch category. Right: 3-D scatter

The PC2/PC3 plane reveals a somewhat circular organisation of epoch categories (Figure 2). An algebraic circle fit (RMSE = 5.616) shows the four epoch groups occupying distinct arcs: BC-era prompts cluster on the left arc, ancient AD-era prompts on the right, modern prompts at the top, and medieval through early modern prompts at the bottom. This suggests the PC2/PC3 plane encodes additional temporal structure beyond the PC1 axis, with epoch categories occupying geometrically distinct regions, though the high circle fit error (RMSE = 5.616) indicates the circular structure is approximate rather than precise.

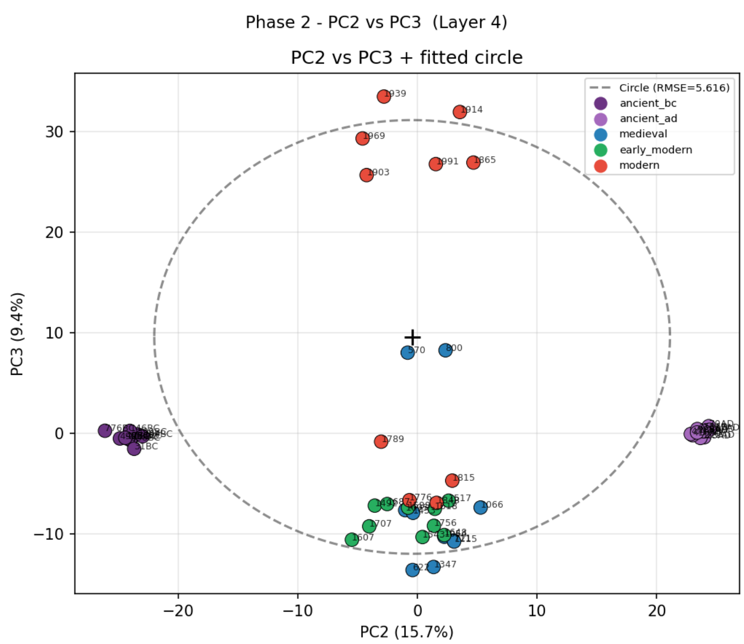

Figure 2: PC2 vs PC3 at layer 4 with algebraic circle fit (RMSE = 5.616).

Angular position on the fitted circle correlates moderately with numeric year ($r = +0.511$, Figure 3), with epoch categories occupying discrete angular bands rather than a smooth progression, confirming the circular structure reflects epoch-level separation rather than a continuous encoding of chronological position.

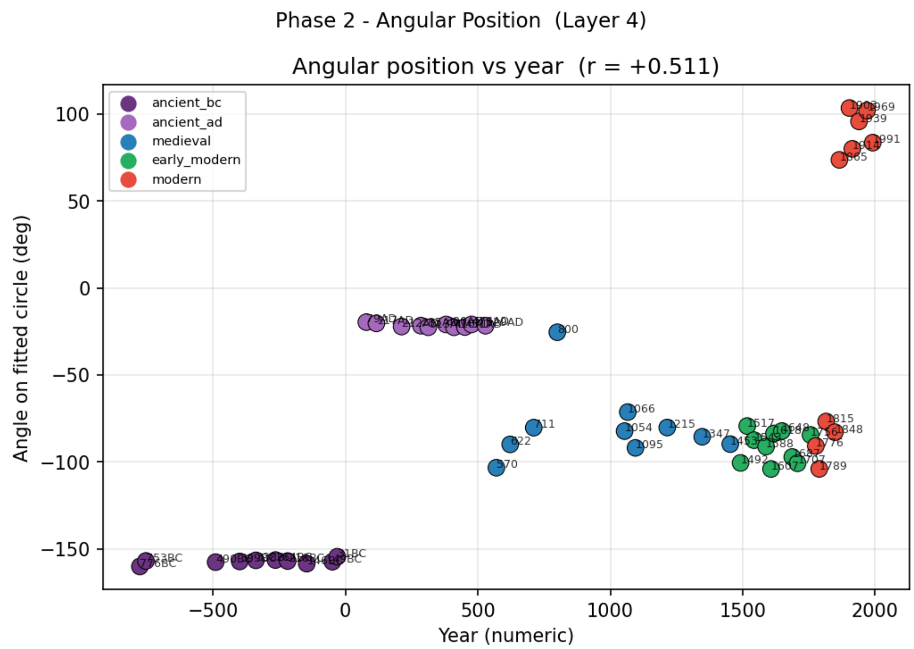

Figure 3: Angular position on fitted circle vs numeric year at layer 4 ($r = +0.511$). Points coloured by epoch category.

5.2 Superposition of Temporal Representations Across Depth

The superposition argument rests on reading $|r|_{\text{base}}$ and $\max_i \Delta_i$ together. A high $|r|_{\text{base}}$ with low $\max_i \Delta_i$ is the fingerprint of distributed superposition — the temporal signal is present and SAE-recoverable, but cannot be attributed to any single feature, as the remaining features compensate collectively for any individual ablation. Conversely, a high $|r|_{\text{base}}$ with high $\max_i \Delta_i$ indicates a concentrated representation where one or a few features are individually load-bearing.

Figure 4 shows the layer-wise depth scan for the original BC/AD corpus. The right panel shows $|r|_{\text{base}}$ remaining stable around 0.85–0.92 across layers 0–9, confirming the temporal signal is consistently present throughout the network. The left panel shows that the $\max_i \Delta_i$ starts high at layers 0–1 (≈ 0.43, well above the load-bearing threshold of 0.10), then declines monotonically through layers 2–9, falling below the load-bearing threshold by layer 7. This divergence between flat $|r|_{\text{base}}$ and declining $\max_i \Delta_i$ is the superposition signature: the temporal signal is maintained at constant strength but becomes progressively impossible to attribute to any individual SAE feature.

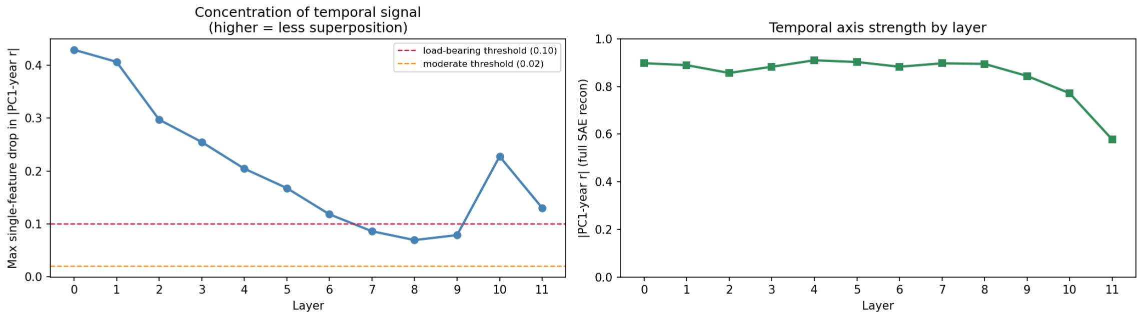

Figure 4: Layer-wise superposition depth scan for the original BC/AD corpus. Left: maximum single-feature ablation drop per layer with load-bearing ($\Delta > 0.10$) and moderate ($\Delta > 0.02$) thresholds. Right: baseline temporal correlation $|r|_{\text{base}}$ per layer.

The full distribution of ablation drops at each layer is shown in Figure 5. At every layer the vast majority of features have negligible drops — the violin bodies are compressed near zero throughout. The upper tails at layers 0–6 represent single outlier load-bearing features, confirming the pattern of one dominant feature with all others negligible rather than a uniform distribution of contributions. From layer 7 onward the tails shrink below the load-bearing threshold, marking the transition to fully distributed superposition.

Two anomalies are worth noting. At layer 10 $\max_i \Delta_i$ spikes back to ≈ 0.23 after falling near zero at layers 7–9; at layer 11 $|r|_{\text{base}}$ drops sharply to ≈ 0.58 and the violin shows a wide body of large negative drops — features whose removal *improves* temporal linearity. Both anomalies occur in the final two layers, consistent with — though not directly evidencing — a reorganisation of residual stream content prior to unembedding.

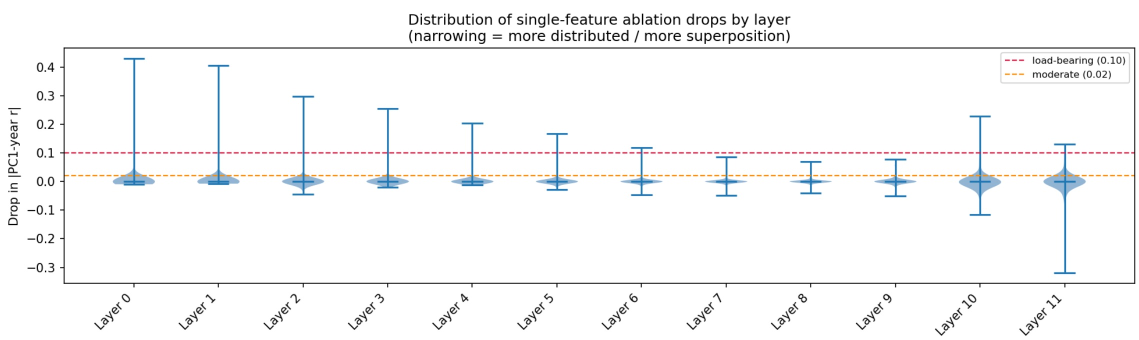

Figure 5: Distribution of single-feature ablation drops by layer for the original BC/AD corpus. Violin bodies compressed near zero at all layers confirm the superposition signature — one outlier load-bearing feature at early layers with all others negligible, transitioning to fully distributed encoding at layers 7–9.

5.3 Surface Token Contamination

Repeating the depth scan on the clean corpus reveals that BC/AD tokens substantially inflate early-layer signals in the original corpus. Figure 6 shows that without BC/AD tokens, PC1 correlation at layer 4 drops from $r = -0.920$ to $r = 0.723$, with PC2 now carrying more temporal information ($r = 0.525$) than before, indicating the temporal signal is more distributed across principal components when surface token identity is unavailable.

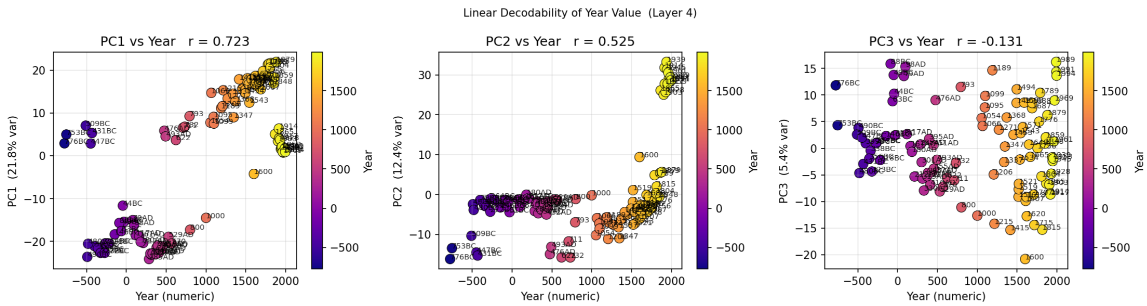

Figure 6: Linear decodability of year value at layer 4 for the clean corpus. PC1 $r = 0.723$, PC2 $r = 0.525$, PC3 $r = -0.131$.

An interesting result is the collapse of $|r|_{\text{base}}$ to near zero at layer 1 (Figure 7, right panel), confirming that the original corpus's strong early-layer signal was driven by surface token detectors rather than semantic content. Genuine temporal encoding builds monotonically from layer 2, reaching ≈ 0.90 by layers 9–11 and remaining stable through the final layer — unlike the original corpus which degraded at layers 10–11. The concentration profile (left panel) peaks at layers 4–5 rather than layer 0, showing that genuinely load-bearing temporal features emerge at mid-layers. From layer 6 onward both corpora converge to the same fully distributed superposition regime.

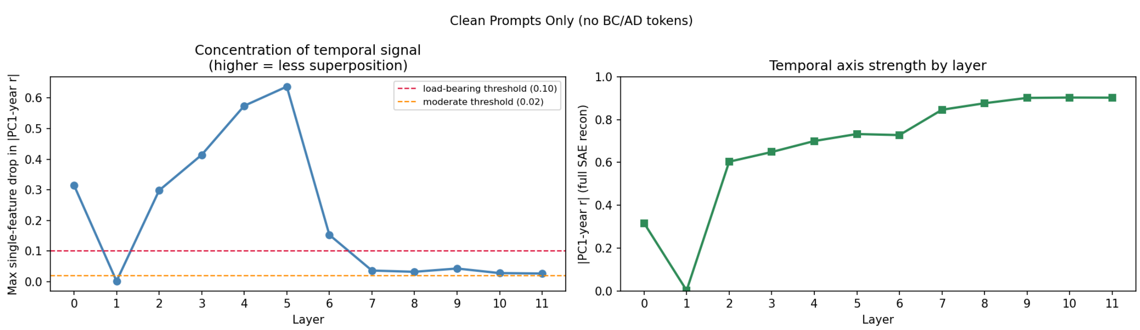

Figure 7: Layer-wise depth scan for the clean corpus. Left: maximum single-feature ablation drop. Right: $|r|_{\text{base}}$ per layer. $|r|_{\text{base}}$ collapses at layer 1 before building monotonically to ≈ 0.90 by layers 9–11.

5.4 Cyclical Concepts: Days of Week and Months of Year

Prior work by Engels et al. [2024] demonstrated that GPT-2 Small encodes days of the week and months of the year as circular representations in the PC2/PC3 plane of the residual stream. We replicate and extend these findings, additionally examining days of the month — a novel cyclic concept with 31 labels that has not previously been studied in this context.

Figure 8 shows the PC2/PC3 geometry for all three concepts at their best layers. Days of the week (layer 7, $r_{\text{angular}} = 0.662$, RMSE = 8.201) and months of the year (layer 7, $r_{\text{angular}} = 0.748$, RMSE = 7.070) both show clear circular organisation, with concepts arranged sequentially around the fitted circle. Days of the month (layer 4, $r_{\text{angular}} = 0.495$, RMSE = 2.651) shows a weaker but present circular structure, with points partially arranged along an arc rather than a full circle. The lower r_{angular} for days of the month is consistent with the increased difficulty of encoding 31 distinct labels compared to 7 or 12.

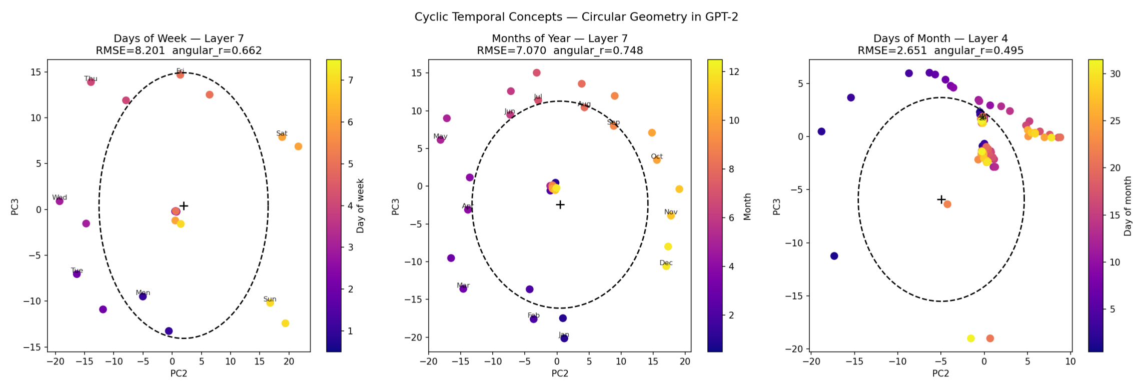

Figure 8: PC2/PC3 geometry for cyclic temporal concepts in GPT-2 Small. Left: days of the week at layer 7 ($r_{\text{angular}} = 0.662$, RMSE = 8.201). Centre: months of the year at layer 7 ($r_{\text{angular}} = 0.748$, RMSE = 7.070). Right: days of the month at layer 4 ($r_{\text{angular}} = 0.495$, RMSE = 2.651). Colour indicates the numeric label for each concept.

5.5 Circular Encoding of Days of the Month

Figure 9 shows the PC2/PC3 geometry for days of the month across five layers and three prompt frames. The all-frames rows reveal that the circular structure is weak and prompt-dependent — no layer produces a clean circular arrangement when all three frames are combined.

Examining frames individually tells a clearer story. Frame 1 (simple statement: *"Today is the Nth of the month"*) collapses to a tight cluster near the origin at all layers, carrying almost no day-number information in the PC2/PC3 plane. Frame 2 (wraparound sequential: *"The day after the Nth is the (N+1)th"*) produces the strongest circular structure, with points arranged along a partial arc that becomes more linear at layers 6–8. Frame 3 (positional: *"The Nth is the Nth day of the month"*) shows linear rather than circular organisation, with day number varying along a single axis rather than wrapping around a circle.

The frame dependence suggests the model does not maintain a single unified circular representation of days of the month. Instead, different prompt framings engage different representational structures.

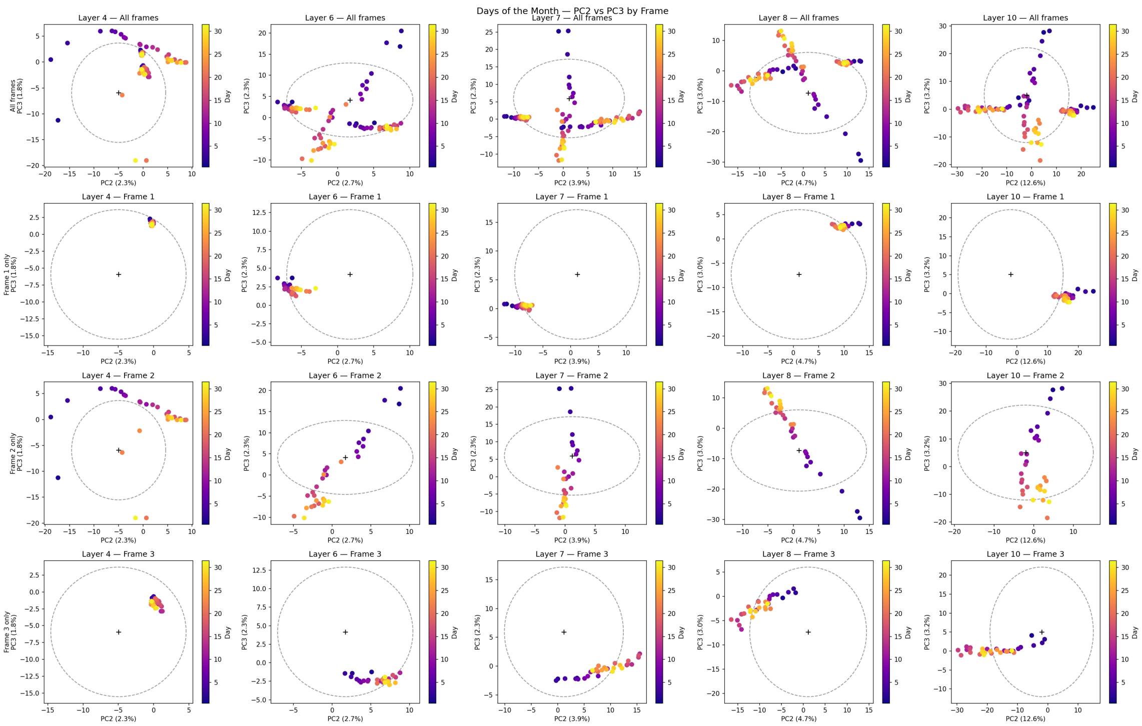

Figure 9: PC2/PC3 geometry for days of the month across five layers (columns) and three prompt frames (rows 2–4), with all frames combined in row 1. Colour indicates day number (1–31).

5.6 Superposition of Circular Representations Across Depth

To examine whether the circular representations identified in Section 5.4 are concentrated in individual SAE features or distributed across many, we repeat the layer-wise ablation scan of Section 5.2 for days of the week and months of the year, replacing the linear $|r|_{\text{base}}$ metric with r_{angular} throughout.

Figure 10 shows the results across all 12 layers. The two concepts exhibit different depth trajectories. Days of the week begin with a moderate circular signal at layer 0 ($r_{\text{angular}} = 0.768$) that persists through layers 1–10 before collapsing entirely at layer 11 ($r_{\text{angular}} = 0.266$). Months of the year follow the opposite pattern: the circular signal starts weaker ($r_{\text{angular}} = 0.534$ at layer 0), builds gradually through the network, and peaks at layer 11 ($r_{\text{angular}} = 0.793$). This crossover suggests the two concepts are built at different stages of processing — days of the week are encoded early and partially dismantled in the final layer, while months of the year are constructed progressively and reach peak clarity only at the output.

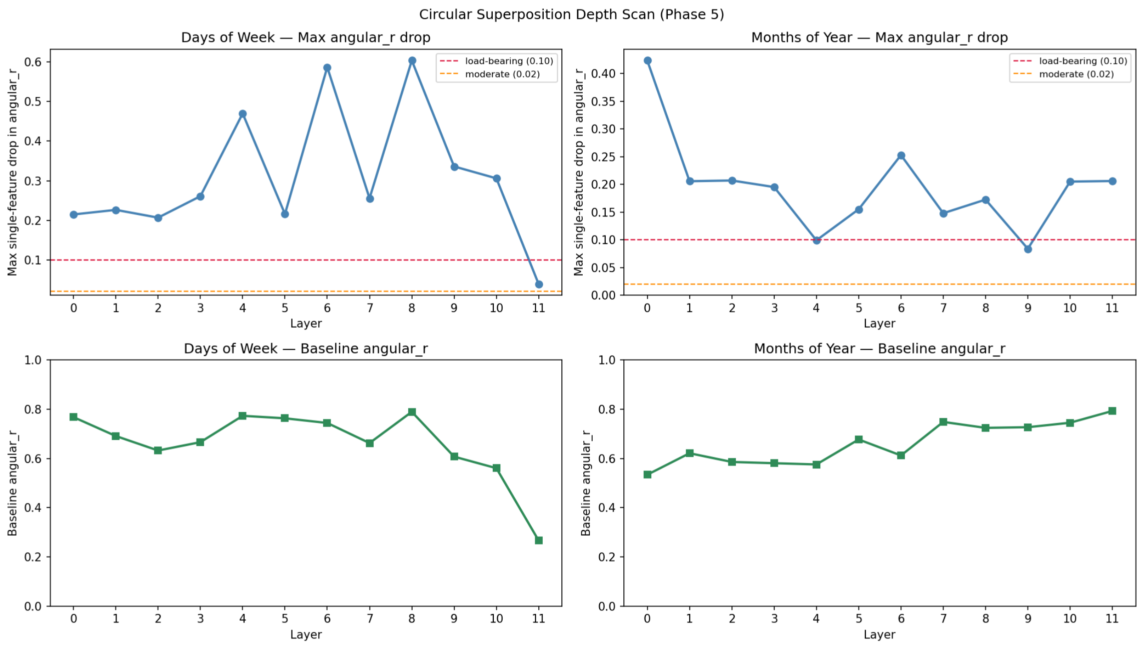

Figure 10: Layer-wise circular superposition scan for days of the week (left) and months of the year (right). Top row: maximum single-feature ablation drop in r_{angular} per layer. Bottom row: baseline r_{angular} per layer.

The concentration profiles (top row) contrast equally sharply with the linear temporal case. For linear time, max single-feature drop declines monotonically from ≈ 0.43 to near zero, indicating a clean transition to distributed superposition. For cyclic concepts, the drop oscillates irregularly throughout the network with no stable convergence. Days of the week show spikes at layers 4 (0.47), 6 (0.59), and 8 (0.60), interspersed with lower-concentration layers — suggesting the circular signal is repeatedly re-concentrated into individual features and then dispersed, rather than handed off cleanly. Months of the year show a high spike at layer 0 (0.42) followed by a broadly lower profile, with no layer crossing the load-bearing threshold of 0.10 between layers 4 and 9, before reconcentrating at layers 10–11. These oscillating patterns are inconsistent with the monotonic superposition lifecycle observed for linear time and suggest that cyclic representations require active, layer-by-layer maintenance rather than a single construction phase followed by passive distribution.

6 Discussion

6.1 Linear vs Circular Temporal Encoding

The most striking contrast between linear and cyclic temporal concepts is the shape of their concentration profiles across depth. For historical time, max single-feature drop declines monotonically from ≈ 0.43 at layer 0 to near zero by layer 7, indicating a clean hand-off from concentrated to distributed encoding. For days of the week and months of the year, the concentration profile oscillates with no stable convergence — individual features become load-bearing at some layers and not others, repeatedly across depth.

One possible interpretation is that maintaining a circular geometric constraint is more demanding than maintaining a linear axis. A linear temporal representation can be encoded as a single direction and passively preserved by residual connections; a circular representation requires equidistant angular spacing across all labels simultaneously, which may need to be actively re-imposed at multiple layers as other computations modify the residual stream. We offer this as a hypothesis rather than a conclusion, as directly testing it would require intervention experiments beyond the scope of this work.

The contrast between days of the week and months of the year is also notable. Days are encoded strongly from layer 0 and collapse at layer 11; months build gradually and peak at layer 11. Whether this reflects differences in training data frequency, the complexity of the underlying concept, or the degree to which each concept is compositionally derived from other representations is an open question.

6.2 The BC/AD Contamination Finding

The collapse of $|r|_{\text{base}}$ to near zero at layer 1 when era markers are removed is a methodological finding with implications beyond this paper. The original corpus's strong layer-0 and layer-1 signal was not evidence of early semantic temporal encoding — it reflected surface token detectors firing on "BC" and "AD". Any probing study on a concept with strong surface-level cues risks the same confound: early-layer probe accuracy can be driven by token identity rather than semantic content.

The clean-corpus results show that genuine semantic encoding of temporal order builds from layer 2 onward, reaching comparable strength to the original corpus by layer 6 and remaining stable through layer 11. This suggests the model does eventually construct a robust temporal representation independent of surface markers, but that construction is masked by the shortcut in corpora that include them.

6.3 Days of the Month

Days of the month extend the circular encoding analysis to a concept with 31 labels, substantially more than days of the week (7) or months of the year (12), and one that appears less frequently in natural text in explicitly cyclic contexts. Despite this, the model encodes a detectable circular geometry at layer 4 ($r_{\text{angular}} = 0.495$, RMSE $= 2.651$), suggesting the circular encoding mechanism is not limited to well-studied, high-frequency cyclic concepts. The weaker signal relative to days of the week and months of the year is consistent with sparser training signal, though we cannot rule out that 31 labels simply produce a noisier circular fit given our dataset size.

6.4 Limitations

The historical year corpus contains only 100 prompts across five epoch categories (20 per group), which is underpowered for ANOVA-based feature selection and may allow noise features into the candidate set at layers where temporal signal is weak. All experiments are conducted on GPT-2 Small; whether the superposition lifecycle, BC/AD contamination effect, or contrasting cyclic depth trajectories generalise to larger or differently trained models is unknown.

Our metrics rest on several geometric assumptions. The linear temporal metric assumes the temporal axis is the dominant source of variance and therefore aligns with PC1; at layers where other signals dominate, this may understate true temporal encoding strength. Circular structure is assumed to occupy the PC2/PC3 plane, so encodings that live in a different component pair at some layers would be missed. Restricting ablation to the top-100 features by ANOVA F-score means broadly-firing load-bearing features — those that score low on ANOVA despite being causally relevant — could be excluded from the candidate set entirely.

All metrics are measured in SAE reconstruction space rather than the true residual stream. SAEs do not perfectly reconstruct their input, so temporal structure not captured by the feature basis is invisible to our analysis. Additionally, late-layer metric degradation at layers 10–11 may partly reflect higher SAE reconstruction error at those layers rather than a genuine weakening of the temporal representation; we do not report per-layer reconstruction error.

7 Conclusion

We investigated how GPT-2 Small encodes temporal concepts in its residual stream using PCA and a layer-wise SAE feature ablation scan. For linear historical time, the temporal axis is consistently decodable across all 12 layers, but transitions from concentrated to distributed superposition with depth: a single SAE feature is load-bearing at early layers, while no individual feature is load-bearing by layer 7. Removing BC/AD surface tokens collapses the early-layer signal to near zero, revealing that the strong layer-0–1 signal in the original corpus reflects surface token detection rather than semantic temporal encoding. Genuine temporal encoding builds from layer 2 onward and converges to the same distributed regime regardless of whether era markers are present.

For cyclic concepts, we replicate circular PC2/PC3 geometry for days of the week and months of the year, and extend the analysis to days of the month, a 31-label concept not previously studied in this context. A layer-wise ablation scan reveals that the two well-established cyclic concepts follow opposite depth trajectories — days of the week encoded strongly from early layers and collapsing at layer 11, months building progressively to peak at layer 11 — and that neither undergoes the monotonic superposition lifecycle observed for linear time. The oscillating concentration profiles suggest cyclic representations require active maintenance across layers rather than a single construction phase.

Future work should examine whether these patterns generalise across model scales and architectures, test whether causal interventions on individual SAE features can manipulate temporal behaviour in model outputs, and investigate the mechanism behind the contrasting depth trajectories for days of the week and months of the year. Alternative feature selection methods — such as mutual information, sparse regression, or activation patching — should also be explored, as ANOVA F-score selects for features that discriminate between predefined group labels and may miss features that encode temporal structure in other ways.

Author Contributions

Sam Brown designed and implemented the experimental pipeline, including the SAE feature ablation scan, cyclic concept datasets, and all six analysis phases. Peyton Calvert contributed to prompt design, result interpretation, and writeup. Both authors contributed to the overall project direction and final paper.

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A Additional Figures

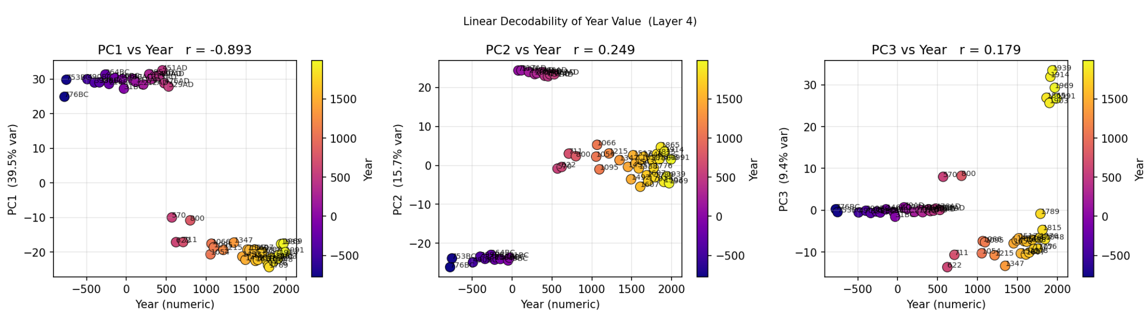

Figure 11: Linear decodability of year value at layer 4 for the original BC/AD corpus. PC scores plotted against numeric year for the top three principal components.

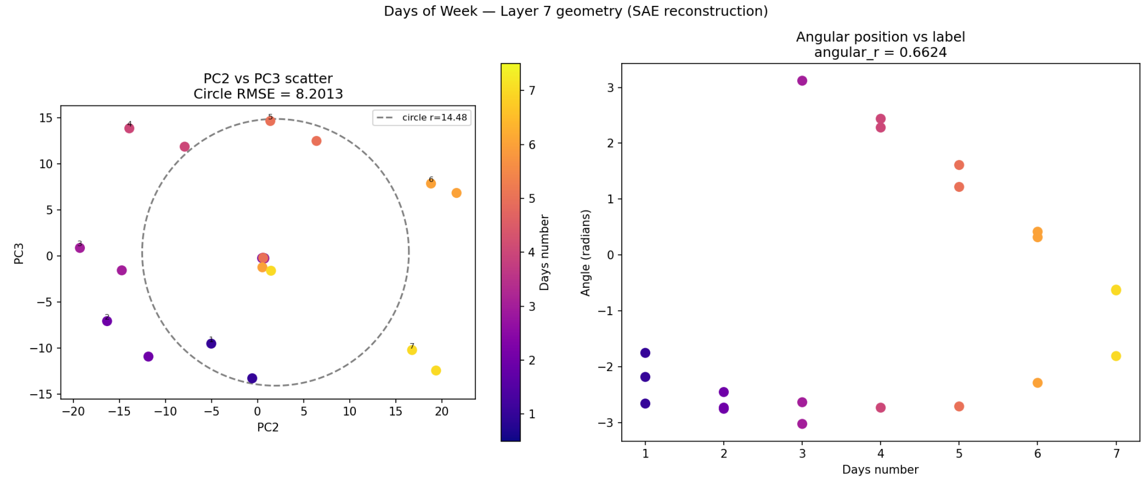

Figure 12: PC2/PC3 circular geometry for days of the week at layer 7 ($r_{\text{angular}} = 0.662$). Left: PC2 vs PC3 scatter with fitted circle. Right: angular position vs day label.

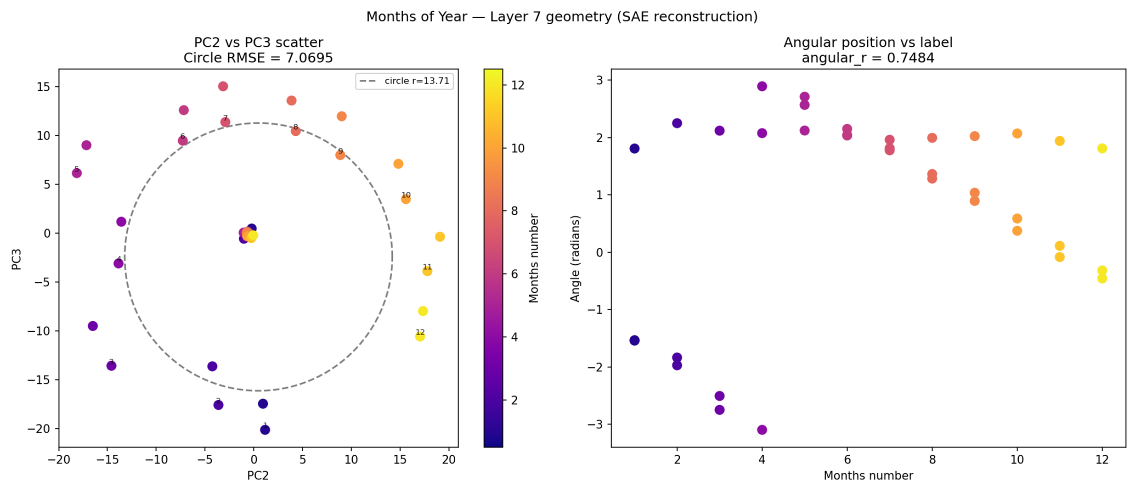

Figure 13: PC2/PC3 circular geometry for months of the year at layer 7 ($r_{\text{angular}} = 0.748$). Left: PC2 vs PC3 scatter with fitted circle. Right: angular position vs month label.

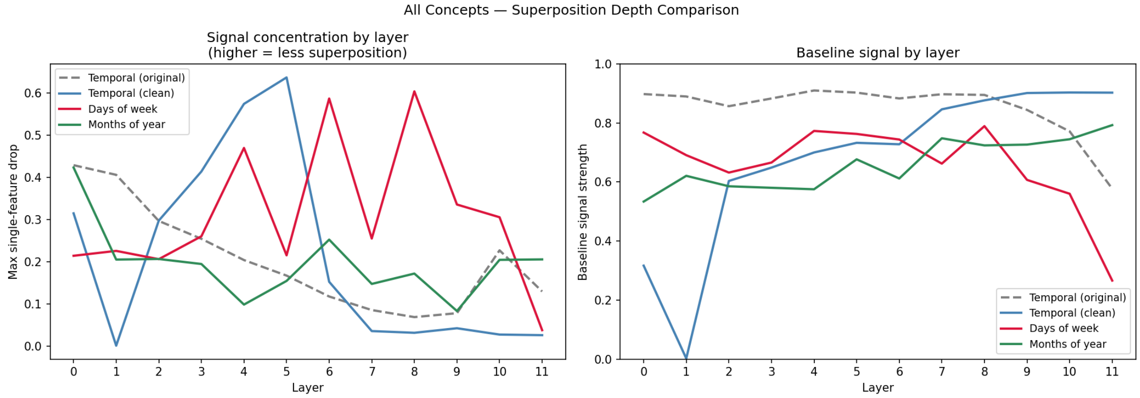

Figure 14: PC2/PC3 geometry comparison across all three cyclic concepts at layer 7.

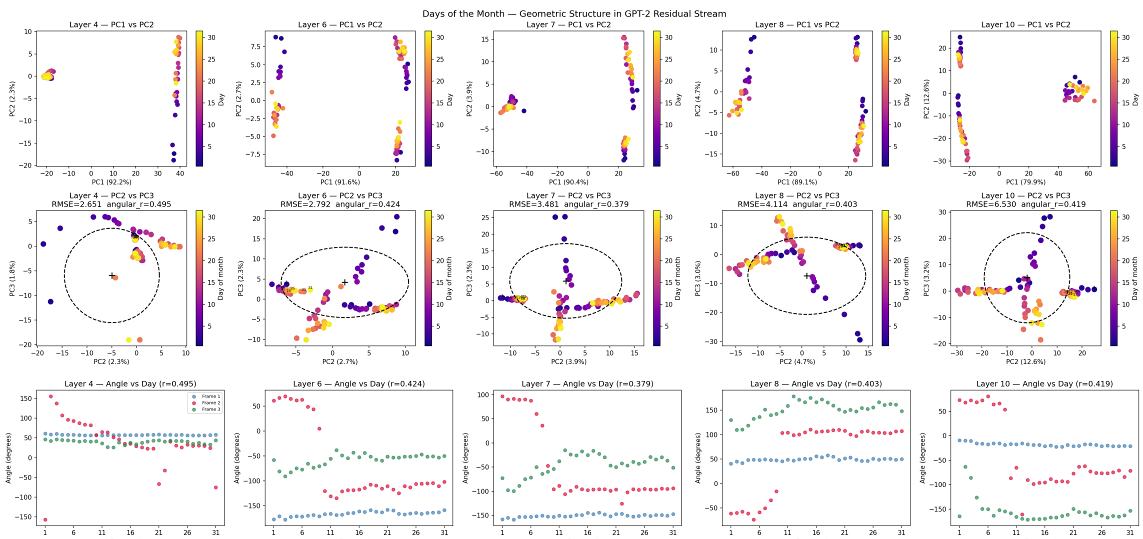

Figure 15: PC2/PC3 circular geometry for days of the month (all frames combined) across selected layers.

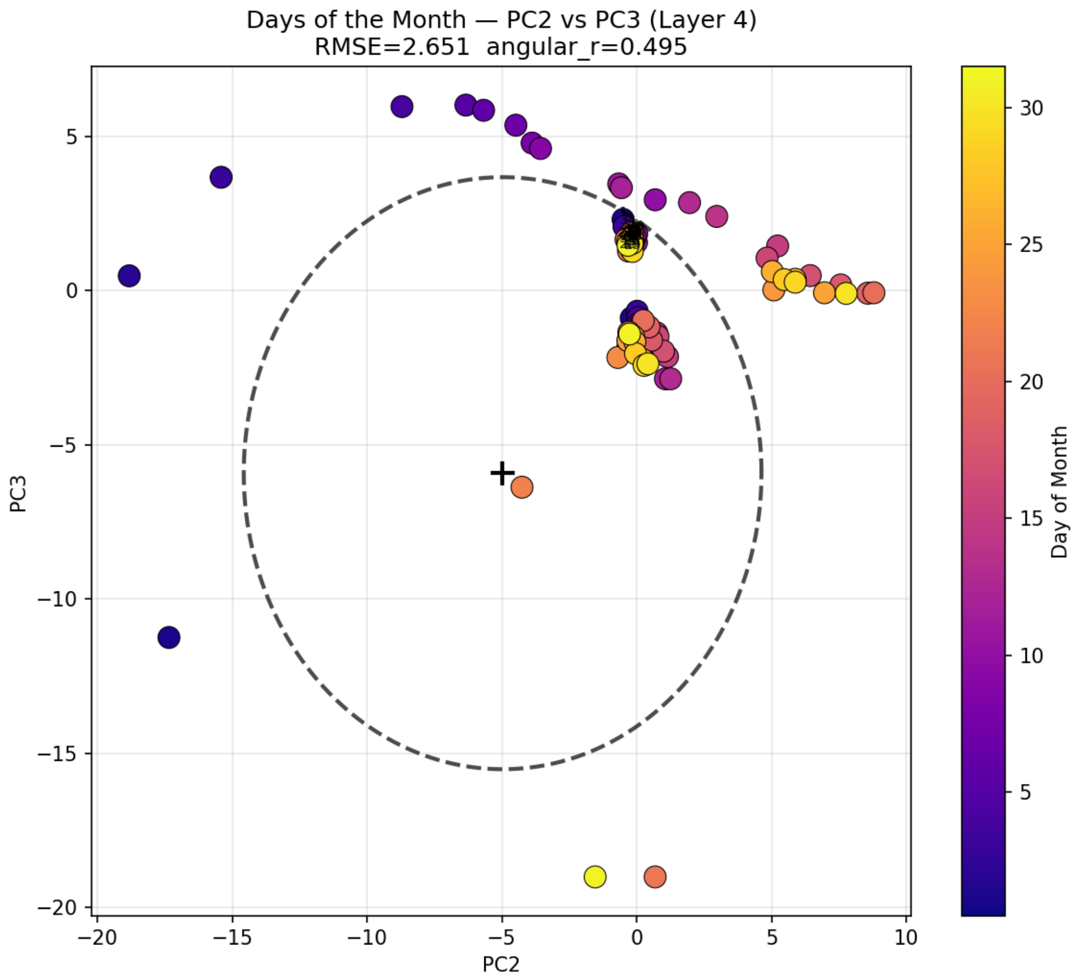

Figure 16: PC2/PC3 geometry for days of the month at the best-performing layer ($r_{\text{angular}} = 0.495$, layer 4).